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IEGP

SPARE PHILOSOPHY

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PROJECT NAME: PMC SERVICES FOR ETHYLENE GLYCOL PROJECT AT PARADIP REFINERY

TOYO JOB NO. : 6351

OWNER: INDIAN OIL CORPORATION LIMITED.

PMC : TOYO ENGINEERING INDIA PRIVATE LIMITED.

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1) Introduction:

The purpose of this document is to define the CONNTRACTOR's responsibilities to provide spare parts. It provides OWNER's minimum requirements for the Plant.

This Specification covers the selection for the following:

- MANDATORY SPARES
- ERECTION & COMMISSIONING SPARES
- TWO YEARS OPERATION AND MAINTENANCE SPARES
- SPECIAL TOOLS

Spare parts will be required for all Mechanical equipment's, piping items, Fire Fighting, Electrical and Instrument items for the Plant.

Throughout this Specification, the term "spare parts" is used in a broad sense, i.e. the component parts of any piece of equipment.

2) DEFINITIONS AND ABBREVIATIONS

For the purposes of this Specification, the following definitions shall apply:

- **OWNER** : Means the firm, INDIAN OIL CORPORATION LTD
- **Project Management Consultant (PMC)**: Means the firm, Toyo Engineering India Pvt. Ltd.
- **Equipment / Services :** Means the equipment and/or services, (including spare parts) including but not limited to, materials, equipment, fabricated products, software, drawings, certification and other documentation, to be supplied under the Contract.
- **Project**: Means IOCL's, Ethylene Glycol Project, Paradip, Odisa, India.
- **Plant** : Means the IOCL's, Ethylene Glycol Project, Paradip, Odhisa, India in connection with the construction and/or operation of which the Equipment / Services are to be supplied.
- **Shall and Must**: Indicate a mandatory requirement.

- **Mandatory Spares:** Parts of equipment, equipment assemblies, or complete items of equipment that will be required for replacement of items not subject to deterioration by normal use but where failure of which will be critical for continuous safe operation of the plant.

  These spares shall be available on site prior to equipment start-up. Where required it is necessary for Mandatory Spare Parts to be manufactured at the same time as the equipment, i.e., where fitting of the part into the equipment is necessary during manufacture.

- **Erection and Commissioning Spares**: Spare parts needed to adequately cover the requirements of day to day maintenance and any premature failures for the period of start-up, commissioning and trial run operations. Commissioning Spare Parts shall be ordered at such a date that they should arrive on site at the same time as the parent equipment, but in any case prior to the equipment commissioning and start-up activities. Commissioning Spare Parts also include parts and assemblies normally used or consumed during installation of equipment, cleaning, flushing, and inspection.

- **Two Years Operation and Maintenance Spares**: Parts or assemblies normally used or consumed on the basis of scheduled maintenance, overhauls, inspections, wear, corrosion, erosion or deterioration in normal service for a period of TWO years after start-up.

- **Special Tools**: These are the tools, jigs and fixtures, including hand held and portable spot measurement instruments which are necessary to carry out maintenance activities on all parts of the Plant.

3) <u>Spare part philosophy:</u>

The Bidder is required to propose innovative means to reduce the necessity for OWNER to maintain quantities of spare parts inventory during the operational life of the equipment. These should include but not be limited to the Bidders holding spare parts on consignment in a central repository for draw down as necessary by and other projects/plants in operation in the region of the project location.

All Erection, Commissioning & Construction Spares, Mandatory Spares and Special Tools, if any, shall be quoted by the Bidder. These spares and tools shall be purchased along with the main equipment.

Bidder shall also provide price list of two years Normal Operation & Maintenance Spares, which shall be ordered separately. All spares and tools except for of two years Normal Operation & Maintenance Spares will be used for price comparison purpose. An Un-priced list of spares and tools shall be enclosed with the Un-Priced Bid.

4) Pricing:

Quoted prices shall be inclusive of Mandatory Spares and Erection / Commissioning spares and special tools and tackles, as per the tender documents.

Bidder's shall separately quote item wise unit price and quantity of recommended spares for 2 years operation. This price shall not be considered for evaluation.

5) The following Sections describe the minimum mandatory spare philosophy to be adopted:

Section A: Mechanical – Static

A.1 Heat Exchangers / Air Cooler

ITEM	SPARE PART	QUANTITY REQUIRED (BASED ON THE NUMBER OF INSTALLED ITEMS)
SHELL AND TUBE EXCHANGERS		
1	Gaskets (each type)	400%
2	Bolts/studs and nuts for nozzles with companion flanges.	10% (Min 2 Sets)
3	Tube sheet girth joints of each size/type	10%
4	Test Ring/ Test Flange/ Dummy Shell / any other accessory required for hydro test	1 Set For Each Exchanger/Cooler/ Condenser (Irrespective of interchangeability)
AIR COOLERS		
1	Gaskets for Header Cover and nozzles	400%
2	Gaskets for Plugs	200%
3	Plugs	20%
4	Bolts/studs and nuts	10% (Min 2 Sets)
5	Belts and Bearings	100%
6	Fan blades for each size/type of Fan	Minimum 1 set of blade, 1 pair of Fan-motor pulley, 1 No blade Hub, 1 no complete bearing housing assembly.
7	Diaphragm Actuator /Positioner for each type of Auto- Variable Pitch Fan	1 set of Diaphragm Actuator with positioner for each type of auto variable pitch fan.

A.2 Columns/Reactors/Vessels/Tanks/Trays & Tower Internals

ITEM	SPARE PART	QUANTITY REQUIRED (BASED ON THE NUMBER OF INSTALLED ITEMS)
COLUMNS/REACTORS/VESSELS/TANKS		
1	Bolting for each nozzle with blind companion flange of each size/type	10% (Min 2 Sets)
2	Gaskets for each nozzle with blind companion flange of each size/type	200%
3	Bolting for internal flanges of each size/type	10% (Min 2 Sets)
4	Gaskets for internal flanges flange of each size/type	200%
5	Demister Pads	100%
TRAYS AND TOWER INTERNALS		
1	Bolts/nuts of each size/type	10% (Min 2 Sets)
2	Clamp Assemblies	10%
3	Valves for Trays	5%
4	Gasketing or tapes	100%
5	Sealing Foil for Cartridge Trays	200%
6	U-Clamps	10%
7	Tower Packings	10%-Metallic / Plastic 15%-Carbon/Ceramic

Section B: Mechanical – Rotary

B.1 Centrifugal Pump (Horizontal / Inline- General Purpose Process)

ITEM	SPARE PART	Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	MECHANICAL								1,2,3
	Pump								
1	Set of Impellers (Full Dia)	1	1	1	1	1	2	2	
2	Set of Shaft with keys	1	1	1	1	1	2	2	
3	Set of Interstage Shaft sleeves	1	1	1	1	1	2	2	
4	Set of Case wearing rings	1	1	2	3	3	3	3	
5	Set of Impeller wearing rings	1	1	2	3	3	3	3	
6	Set of Gaskets, O-rings	2	4	6	8	8	8	8	
7	Set of Interstage Bushings	1	1	1	1	1	2	2	
8	Set of Mechanical Seals along with Shaft Sleeves under seals with gland plates for Cartridge seals / without gland plate for other seals	1	1	2	2	3	3	3	
9	Set of gland packings	2	4	6	8	8	8	8	
10	Set of Labyrinths for bearing housing	1	1	2	3	3	3	3	
11	Set of Oil Seals for bearing housing	1	2	3	4	4	4	4	
12	Set of bearings (Non drive end side)	1	1	2	3	3	3	3	
13	Set of bearings (Drive end side)	1	1	2	3	3	3	3	
14	Set of flexible element of coupling	1	1	1	1	1	2	2	
15	Set of Bearing Lock Washer	1	1	2	2	2	2	2	
16	Throat Bush	1	1	1	2	2	2	2	
17	Bearing Nut	1	1	1	1	1	1	1	
18	Impeller Nut with washer	1	1	1	1	1	1	1	
19	Deflector	1	1	1	1	1	1	1	
20	Bearing Bracket	1	1	1	1	1	1	1	
21	Bearing Cover	1	1	1	1	1	1	1	
22	Coupling Complete Set (including Fasteners)	1	1	1	1	1	1	1	

23	Set of Mech. Seal Spares (NOTE 4)	1	1	2	2	2	2	2	
	ELECTRICAL								
	LV motor (3Ph Motors >1 KW & <=160KW)								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block (including Terminal Box Cover with Screws)	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	
	HV (6.6/11KV) induction motors >160KW								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block / HT bushing	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	

B.2 Centrifugal Pump (Vertical)

ITEM	SPARE PART	Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)							
		1	2	3	4	5	6	7	Notes
	MECHANICAL								
	Pump								1,2,3
1	Set of Impellers (Full Dia)	1	1	1	1	1	2	2	
2	Set of Pump Shaft with keys (excluding driver & column shaft)	1	1	1	1	1	2	2	
3	Set of Interstage Shaft sleeves	1	1	1	1	1	2	2	
4	Set of Interstage Bushings	1	1	1	1	1	2	2	
5	Set of Case wearing rings	1	1	2	3	3	3	3	
6	Set of Impeller wearing rings	1	1	2	3	3	3	3	
7	Set of Gaskets, O-rings	2	4	6	8	8	8	8	
8	Set of Line Bearings (Spider Bearings)	1	1	2	3	3	3	3	
9	Set of Mechanical Seals along with Shaft Sleeves under seals with gland plates for Cartridge seals/ without gland plate for other seals	1	1	2	2	3	3	3	
10	Set of gland packings	2	4	6	8	8	8	8	
11	Set of Vapour Seals	1	2	3	4	4	4	4	
12	Set of bearings	1	1	2	3	3	3	3	
13	Set of flexible element for coupling	1	1	1	1	1	2	2	
14	Stuffing Box Spacer	1	1	1	1	1	1	1	
15	Set of Mechanical Seal Spares (Note 4)	1	1	2	2	2	2	2	
16	Impeller Nut with Washer	1	1	1	1	1	1	1	
17	Bearing Nut	1	1	1	1	1	1	1	
18	Bearing Bracket	1	1	1	1	1	1	1	
19	Bearing Cover	1	1	1	1	1	1	1	
20	Set of joint rings	1	1	1	1	1	1	1	
21	Coupling Complete Set (including Fasteners)	1	1	1	1	1	1	1	

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	ELECTRICAL								
	LV motor (3Ph Motors >1 KW & <=160KW)								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block (including Terminal Box Cover with Screws)	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	
	HV (6.6/11KV) induction motors >160KW								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block / HT bushing	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	

B.3 Sundyne Pump:

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	MECHANICAL								1,2,3
	Pump								
1	Impeller (Rated Dia.)	1	1	1	1	1	2	2	
2	Set of High Speed Shaft Assy. with Impeller keys	1	1	1	1	1	2	2	
3	Ball bearing for Low Speed Shaft (lower)	1	1	2	2	3	3	3	
4	Ball bearing for Low Speed Shaft (upper)	1	1	2	2	3	3	3	
5	Journal bearing for High Speed Shaft (lower)	1	1	2	2	3	3	3	
6	Journal bearing for High Speed Shaft (upper)	1	1	2	2	3	3	3	
7	Sets of O-ring repair kit	2	4	6	8	8	8	8	
8	Mechanical Seal for Gear Box Side	1	1	2	2	3	3	3	
9	Seal Rotating Face for Gear Box Side	1	1	2	2	3	3	3	
10	Coupling Complete Set (including Fasteners)when applicable	1	1	1	1	1	2	2	
11	Seal rotating Face for process Side	1	1	2	2	3	3	3	
12	Oil Filter	2	4	6	8	8	8	8	
13	Shaft Seal	2	4	6	8	8	8	8	
14	Impeller Tab Washer	1	2	3	4	5	7	7	
15	Low Speed Shaft Assembly (Intermediate/Idler)	1	1	1	1	2	2	2	
16	Shaft Spacer	1	1	1	1	2	2	2	
17	Set of Gaskets	2	2	2	2	3	3	3	
	Addnl. Spares with SINGLE Seal								
1	Set of throttle bushing	1	1	1	1	1	2	2	
2	Shaft sleeve	1	1	2	2	3	3	3	
3	Set of Mech. Seal (Complete Assy., Process Side only)	1	1	2	2	3	3	3	
4	Set of Mech. Seal Parts (as Mech. Seal Repair Kit for Process Side Only)	1	1	2	2	3	3	3	
	Addnl. spares with DOUBLE/TANDEM Seal								
1	Shaft sleeve (lower)	1	1	2	2	3	3	3	
2	Shaft sleeve (upper)	1	1	2	2	3	3	3	
3	Set of Mech. Seal (Complete Assy., Process Lower Side)	1	1	2	2	3	3	3	
4	Set of Mech. Seal (Complete Assy., Process Upper Side)	1	1	2	2	3	3	3	
5	Seal Rotating Face Process Lower side	for	1	1	2	2	3	3	3
6	Seal Rotating Face Process Upper side	for	1	1	2	2	3	3	3

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
7	Set of Mechanical Seal Parts (as Mech. Seal Repair Kit for Process Lower Side Only)	1	1	2	2	3	3	3	
8	Set of Mechanical Seal Parts (as Mech. Seal Repair Kit for Process Upper Side Only)	1	1	2	2	3	3	3	
	ELECTRICAL								
	Motor								
	LV motor (3Ph Motors >1 KW & <=160KW)								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block (including Terminal Box Cover with Screws)	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	
	HV (6.6/11KV) induction motors >160KW								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block / HT bushing	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	

B.4 Centrifugal Horizontal Pumps (Multistage)

ITEM	SPARE PART	Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	MECHANICAL								1,2,3
	Pump								
1	Set of case wearing rings	1	1	2	3	3	3	3	
2	Set of Impeller wearing rings	1	1	2	3	3	3	3	
3	Set of Intermediate bushings	1	1	2	3	3	3	3	
4	Set of Balance Drum Sleave (Bushing)	1	1	2	3	3	3	3	
5	Set of Mechanical Seals along with Shaft Sleeves under seals with gland plates for Cartridge seals without gland plate for other seals	1	1	2	2	3	3	3	
6	Set of Labyrinths for bearing housing	1	1	2	3	3	3	3	
7	Set of Gaskets, O-rings	2	4	6	8	8	8	8	
8	Set of Bearings, NDE	1	1	2	3	3	3	3	
9	Set of Bearings, DE	1	1	2	3	3	3	3	
10	Set of Flexible elements of Coupling	1	1	1	1	1	2	2	
11	Complete Set of Mechanical Seal Spares (Note 4)	1	1	2	2	2	2	2	
12	Throat Bush	1	1	1	2	2	2	2	
13	Lock Nut	1	1	1	2	2	2	2	
14	Deflector	1	1	1	1	1	1	1	
15	Set of impellers	1	1	2	2	2	2	2	
16	Coupling Elements- Complete Set (including Fasteners)	1	2	2	2	2	2	2	
	Additional Spares for Axially Split Pumps								
1	Complete Rotor assembly dynamically balanced including balance drum	1	1	1	1	1	2	2	
2	Set of Interstage bushings	1	1	1	1	1	2	2	
3	Set of Balance Drum Sleeve (Bushing)	1	1	1	1	1	2	2	
	Additional Spares for Radially Split Double Case Pumps								
1	Complete replaceable innercase element consisting of all rotating & stationary hydraulic elements with permanent storage container including necessary fixture to store the rotor in vertical position	1	1	1	1	1	2	2	
	Lube Oil Pump Rotary (Main + Stand By)								
1	Complete bare pump assembly excluding driver couplings & base plate	1	1	1	1	1	2	2	
2	Set of bearings	1	1	2	3	3	3	3	
3	Set of Mechanical Seals	1	1	2	2	3	3	3	
4	Set of Gaskets , o-rings	2	4	6	8	8	8	8	

ITEM	SPARE PART	Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	Lube Oil Filters								
1	Filter cartridge / element along with set of gaskets, o-rings	10	20	30	30	30	30	30	
	Lube Oil Coolers								
1	Set of Gaskets, o-rings	2	4	6	8	8	8	8	
	ELECTRICAL								
	LV motor (3Ph Motors >1 KW & <=160KW)								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block (including Terminal Box Cover with Screws)	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	
	HV (6.6/11KV) induction motors >160KW								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block / HT bushing	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	

B.5 Rotary Pump

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	MECHANICAL								1,2,3
	Pumps								
1	Set of Rotors	1	1	1	1	1	2	2	
2	Casing Insert (Stator) [only for screw pump]	1	1	1	1	1	2	2	
3	Set of Bearings (Drive End) - Each Screw	1	1	2	3	3	3	3	
4	Set of Bearings (Non Drive End) - Each Screw	1	1	2	3	3	3	3	
5	Set of Gaskets, O - rings	2	4	6	8	8	8	8	
6	Set of Labyrinths	1	1	2	3	3	3	3	
7	Set of Oil Seals	1	2	3	4	4	4	4	
8	Set of Mechanical Seals along with Shaft Sleeves under seals with gland plates for Cartridge seals / without gland plate for other seals	1	1	2	2	3	3	3	4
9	Set of Gland Packings (Where Applicable)	2	4	6	8	8	8	8	
10	Coupling Complete Set (including Fasteners)	1	1	1	1	1	2	2	
	ELECTRICAL								
	MV motor (3Ph Motors >1 KW & <=160KW)								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block (including Terminal Box Cover with Screws)	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	

B.6 Reciprocating Pump

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	2	3	4	5	6	7	
	MECHANICAL								**Notes**
	Pumps								1,2,3
1	Set of Suction Valve Assembly	1	1	2	2	3	3	3	
2	Set of Discharge Valve Assembly	1	1	2	2	3	3	3	
3	Set of bearings	1	1	2	3	3	3	3	
4	Set of Gaskets, O - rings	2	4	6	8	8	8	8	
5	Set of Plungers	1	1	1	1	1	2	2	
6	Set of Plunger Packings	2	4	6	8	8	8	8	
7	Valve Seat	1	1	2	2	2	2	2	
8	Oil seal	1	2	2	2	3	3	3	
9	Valve Disc	1	1	2	2	2	2	2	
10	Ball	1	1	2	2	2	2	2	
	ELECTRICAL								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block (including Terminal Box Cover with Screws)	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	

B.7 Reciprocating Pump (Diaphragm Type)

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	MECHANICAL								1,2,3
	Pumps								
1	Set of Suction Valve Assembly	1	1	2	2	3	3	3	
2	Set of Discharge Valve Assembly	1	1	2	2	3	3	3	
3	Set of bearings	1	1	2	3	3	3	3	
4	Set of Gaskets, O - rings	2	4	6	8	8	8	8	
5	Set of Diaphragms	2	2	4	6	6	6	6	
6	Set of Hydraulic packings / piston rings	2	2	4	6	6	6	6	
7	Valve Seat	1	1	2	2	2	2	2	
8	Oil Seal	1	2	2	2	3	3	3	
9	Valve Disc	1	1	2	2	2	2	2	
10	Ball	1	1	2	2	2	2	2	
	ELECTRICAL								
	LV motor (3Ph Motors >1 KW & <=160KW)								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block (including Terminal Box Cover with Screws)	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	

B.8 Positive Displacement Pump (Controlled Volume)

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	MECHANICAL								1,2,3
	Pumps								
1	Set of Suction Valve Assembly	1	1	2	2	3	3	3	
2	Set of Discharge Valve Assembly	1	1	2	2	3	3	3	
3	Set of bearings	1	1	2	3	3	3	3	
4	Set of Gaskets, O - rings	2	4	6	8	8	8	8	
5	Set of oil seals	1	1	2	2	2	3	3	
	Additional Spares for Plunger Pumps								
1	Set of Plungers	1	1	1	1	1	2	2	
2	Set of Plunger Packings	2	4	6	8	8	8	8	
3	Worm Wheel Assembly	1	1	2	2	2	2	2	
	Additional Spares for Diaphragm Pumps								
1	Set of Diaphragms	2	2	4	6	6	6	6	
2	Set of Hydraulic piston packings / rings	2	2	4	6	6	6	6	
3	Worm Wheel Assembly	1	1	2	2	2	2	2	
	ELECTRICAL								
	LV motor (3Ph Motors >1 KW & <=160KW)								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block (including Terminal Box Cover with Screws)	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	

B.9 Reciprocating Compressor:

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	MECHANICAL								1,2,3,4
	Compressor								
1	Set of all Compressor Bearings/bushes (main crank shaft bearings, connecting rod big-end & small end bearings, cross head bearings etc.	1	1	2	3	3	3	3	
2	Connecting rod assembly (without bearings)	1	1	1	1	1	2	2	
3	Set of piston rings for all lubricated cylinders	1	1	2	3	3	3	3	
4	Set of piston rings (sealing rings & rider rings) for all mini- lubricated/non-lubricated cylinders	3	3	6	9	9	9	9	
5	Valve assembly (suction) - each type	2	2	4	6	6	6	6	
6	Valve assembly (discharge) - each type	2	2	4	6	6	6	6	
7	Set of Valve springs, Valve etc of each installed type	1	2	3	4	4	4	4	
8	Piston rod packing assembly - each type	1	1	1	1	1	2	2	
9	Set of replaceable part of piston rod packing-lubricated cylinder	1	1	2	3	3	3	3	
10	Set of replaceable part of piston rod packing-non lubricated cylinder	3	3	6	9	9	9	9	
11	Set of all wiper rings	1	1	2	3	3	3	3	
12	Packing cooling oil pump assembly (where applicable)	1	1	1	1	1	2	2	
13	Set of all Gaskets & `O' - rings for the compressor	2	4	6	8	8	8	8	

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
14	Valve unloader assembly - each type	1	1	2	2	3	3	3	
15	Lubricator assembly (full assembly)	1	1	1	1	1	2	2	
16	Set of air filters elements (for air service only)	2	4	6	8	8	8	8	
17	Crankshaft - for Each Type	1	1	1	1	1	1	1	
	Lube Oil Pump Centrifugal (Main + Stand By)								
1	Complete bare pump assembly excluding driver couplings & base plate	1	1	1	1	1	2	2	
2	Set of bearings	1	1	2	3	3	3	3	
3	Set of Mechanical Seals	1	1	2	2	3	3	3	
4	Set of Impeller Wear Rings	1	1	2	3	3	3	3	
5	Set of Casing Wear Rings	1	1	2	3	3	3	3	
6	Set of Gaskets ,'O'- rings	2	4	6	8	8	8	8	
7	Complete set of coupling of each type	1	1	1	1	1	1	1	
	Lube Oil Pump Rotary (Main + Stand By)								
1	Complete base pump assembly excluding driver couplings & base plate	1	1	1	1	1	2	2	
2	Set of bearings	1	1	2	3	3	3	3	
3	Set of Mechanical Seals	1	1	2	2	3	3	3	
4	Set of Gaskets, 'O'- rings	2	4	6	8	8	8	8	

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
5	Complete set of coupling of each type	1	1	1	1	1	1	1	
	Cylinder Cooling Pump Centrifugal (Main + Stand By)								
1	Complete bare pump assembly excluding driver couplings & base plate	1	1	1	1	1	2	2	
2	Set of bearings	1	1	2	3	3	3	3	
3	Set of complete seal with sleeve and gland plate	1	1	2	2	3	3	3	
4	Set of Impeller Wear Rings	1	1	2	3	3	3	3	
5	Set of Casing Wear Rings	1	1	2	3	3	3	3	
6	Set of Gaskets, 'O' - rings	2	4	6	8	8	8	8	
7	Complete set of coupling of each type	1	1	1	1	1	1	1	
	Lube Oil Filters								
1	Filter cartridge / element along with set of gaskets, 'O' - rings	10	20	30	30	30	30	30	
	Lube Oil Coolers								
1	Set of Gaskets, 'O' - Rings	2	4	6	6	8	8	8	
	Interstage Coolers								
1	Set of Gaskets, 'O' - Rings	2	4	6	6	8	8	8	
	After Coolers								
1	Set of Gaskets, 'O' - Rings	2	4	6	6	8	8	8	
	ELECTRICAL								

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	LV motor (3Ph Motors >1 KW & <=160KW)								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block (including Terminal Box Cover with Screws)	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	
	HV (6.6/11KV) induction motors >160KW								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block / HT bushing	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	

B.10 Reciprocating Compressor (Diaphragm Type)

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	MECHANICAL								1,2,3,4
	Compressor								
1	Set of all Compressor Bearings/bushes (main crank shaft bearings, connecting rod big-end & small end bearings, cross head bearings etc.	1	1	2	3	3	3	3	
2	Connecting rod assembly (without bearings)	1	1	1	1	1	2	2	
3	Set of Piston Rings for hydraulic piston/plunger	1	1	2	3	3	3	3	
4	Set of Diaphragms	2	4	6	8	8	8	8	
5	Valve assembly (suction) - each type	2	2	4	6	6	6	6	
6	Valve assembly (discharge) - each type	2	2	4	6	6	6	6	
7	Set of Valve springs, valve plates, etc of each installed type	1	2	3	4	4	4	4	
8	Set of all Gaskets & `O' - rings for the compressor (including plate O-rings, holddown O- rings etc.)	2	4	6	8	8	8	8	
9	Set of Hydraulic Pressure Limiter Device	1	1	1	1	1	1	1	
10	Set of Compensating Pump repair kit	1	1	1	1	1	1	1	
11	Crankshaft - for Each Type	1	1	1	1	1	1	1	
	Lube Oil Pump Rotary (Main + Stand By)								
1	Complete base pump assembly excluding driver couplings & base plate	1	1	1	1	1	2	2	
2	Set of bearings	1	1	2	3	3	3	3	
3	Set of Mechanical Seals	1	1	2	2	3	3	3	
4	Set of Gaskets,'O'- rings	2	4	6	8	8	8	8	
5	Complete set of coupling of each type	1	1	1	1	1	1	1	
	Lube Oil Filters								
1	Filter cartridge / element along with set of gaskets, 'O' - rings	10	20	30	30	30	30	30	
	Lube Oil Coolers								
1	Set of Gaskets, 'O' - Rings	2	4	6	6	8	8	8	
	Interstage Coolers								
1	Set of Gaskets, 'O' - Rings	2	4	6	6	8	8	8	
	After Coolers								
1	Set of Gaskets, 'O' - Rings	2	4	6	6	8	8	8	

	ELECTRICAL								
	LV motor (3Ph Motors >1 KW & <=160KW)								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block (including Terminal Box Cover with Screws)	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	
	HV (6.6/11KV) induction motors >160KW								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block / HT bushing	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	

B.11 Reciprocating Compressor (Labyrinth Type)

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	MECHANICAL								1,2,3,4
	Compressor								
1	Set of all Compressor Bearings/bushes (main crank shaft bearings, connecting rod big-end & small end bearings, cross head bearings etc.	1	1	2	3	3	3	3	
2	Connecting rod assembly (without bearings)	1	1	1	1	1	2	2	
3	Valve assembly (suction) - each type	2	2	4	6	6	6	6	
4	Valve assembly (discharge) - each type	2	2	4	6	6	6	6	
5	Set of Valve springs, Valve plates etc of each installed type	1	2	3	4	4	4	4	
6	Piston rod packing assembly - each type	1	1	1	1	1	2	2	
7	Set of replaceable part of piston rod packing	1	1	2	3	3	3	3	
8	Set of Piston Rod guide bearings	1	1	2	3	3	3	3	
9	Set of Crankshaft seals	1	1	2	2	3	3	3	
10	Set of all Gaskets & `O' - rings for the compressor	2	4	6	8	8	8	8	
11	Valve unloader assembly - each type	1	1	2	2	3	3	3	
12	Crankshaft - for Each Type	1	1	1	1	1	1	1	
	Lube Oil Pump Rotary (Main + Stand By)								
1	Complete base pump assembly excluding driver couplings & base plate	1	1	1	1	1	2	2	
2	Set of bearings	1	1	2	3	3	3	3	
3	Set of Mechanical Seals	1	1	2	2	3	3	3	
4	Set of Gaskets, 'O'- rings	2	4	6	8	8	8	8	

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
5	Complete set of coupling of each type	1	1	1	1	1	1	1	
	Lube Oil Filters								
1	Filter cartridge / element along with set of gaskets, 'O' - rings	10	20	30	30	30	30	30	
	Lube Oil Coolers								
1	Set of Gaskets, 'O' - Rings	2	4	6	6	8	8	8	
	Interstage Coolers								
1	Set of Gaskets, 'O' - Rings	2	4	6	6	8	8	8	
	After Coolers								
1	Set of Gaskets, 'O' - Rings	2	4	6	6	8	8	8	
	Electrical								
	LV motor (3Ph Motors >1 KW & <=160KW)								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block (including Terminal Box Cover with Screws)	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	
	HV (6.6/11KV) induction motors >160KW								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block / HT bushing	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	

B.12 Rotary Compressor (Screw Type)

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	MECHANICAL								1,2,3,4
	Compressor								
1	Set of Screws	1	1	1	1	1	1	1	
2	Set of Radial Bearings (DE + NDE)	1	2	3	4	4	4	4	
3	Set of Thrust Bearings	1	2	3	4	4	4	4	
4	Set of Shaft Seal Assembly (DE + NDE) including sleeves under seals	1	2	3	3	3	3	3	
5	Set of Timing Gears (for oil free screw compressor)	1	1	1	1	1	2	2	
6	Set of Gaskets, O - rings	2	4	6	8	8	8	8	
7	Set of air filters elements	2	4	6	8	8	8	8	
8	Sealing Compounds for casing joints * Suitable for as many applications	4*	8*	8*	8*	8*	8*	8*	
9	Complete Set of Coupling of Each Type	1	1	1	1	1	1	1	
	Lube Oil Pump Centrifugal (Main + Stand By)								
1	Complete bare pump assembly excluding driver couplings & base plate	1	1	1	1	1	2	2	
2	Set of bearings	1	1	2	3	3	3	3	
3	Set of Mechanical Seals	1	1	2	2	3	3	3	
4	Set of Impeller Wear Rings	1	1	2	3	3	3	3	
5	Set of Casing Wear Rings	1	1	2	3	3	3	3	
6	Set of Gaskets, 'O'- rings	2	4	6	8	8	8	8	
7	Complete set of coupling of each type	1	1	1	1	1	1	1	
	Seal Oil Pump Rotary (Main + Stand By)								
1	Complete bare pump assembly including driver couplings & base plate	1	1	1	1	1	2	2	
2	Set of bearings	1	1	2	3	3	3	3	

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
3	Set of Mechanical Seals	1	1	2	2	3	3	3	
4	Set of Gaskets, 'O'- rings	2	4	6	8	8	8	8	
5	Complete set of coupling of each type	1	1	1	1	1	1	1	
	Lube Oil Filters								
1	Filter cartridge / element along with set of gaskets, 'O' - rings	10	20	30	30	30	30	30	
	Seal Oil Filters								
1	Filter cartridge / element along with set of gaskets, 'O' - rings	10	20	30	30	30	30	30	
2	Set of Seal Gas filter elements for Gas Seals	10	20	30	30	30	30	30	
	Oil Coolers								
1	Set of Gaskets, 'O'-rings	2	4	6	8	8	8	8	
	Interstage Coolers								
1	Set of Gaskets, 'O'-rings	2	4	6	8	8	8	8	
	After Coolers								
1	Set of Gaskets, 'O'-rings	2	4	6	8	8	8	8	
	Electrical								
	LV motor (3Ph Motors >1 KW & <=160KW)								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block (including Terminal Box Cover with Screws)	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	HV (6.6/11KV) induction motors >160KW								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block / HT bushing	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	

B.13 Centrifugal Compressor

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	MECHANICAL								1,2,3, 4
1	Complete Rotor dynamically balanced including balance drum (with permanent storage container for vertical storage)	1	1	1	1	1	1	1	
2	Set of Interstage Shaft Sleeves	1	2	3	3	3	3	3	
3	Set of Labyrinths including interstage balance drum & bearing housing	1	2	3	3	3	3	3	
4	Set of Radial Bearings (DE + NDE)	1	2	3	4	4	4	4	
5	Set of Thrust Bearings	1	2	3	4	4	4	4	
6	Set of Shaft Seal Assembly (DE + NDE) including sleeve under seals	1	2	3	3	3	3	3	
7	Set of Gaskets, O - rings	2	4	6	8	8	8	8	
8	Sealing compounds for casing joints (if reqd.) * Suitable for as many applications	4 *	8 *	8 *	8 *	8 *	8 *	8 *	
9	Complete Set Of Coupling Of Each Type	1	1	1	1	1	1	1	
	Lube Oil Pump Centrifugal (Main + Stand By)								
1	Complete bare pump assembly excluding driver couplings & base plate	1	1	1	1	1	2	2	
2	Set of bearings	1	1	2	3	3	3	3	
3	Set of Mechanical Seals	1	1	2	2	3	3	3	
4	Set of Impeller Wear Rings	1	1	2	3	3	3	3	
5	Set of Casing Wear Rings	1	1	2	3	3	3	3	
6	Set of Gaskets, O - rings	2	4	6	8	8	8	8	
7	Complete set of coupling of each type	1	1	1	1	1	1	1	
	Lube Oil Pump Rotary (Main + Stand By)								
1	Complete bare pump assembly excluding driver couplings & base plate	1	1	1	1	1	2	2	
2	Set of bearings	1	1	2	3	3	3	3	
3	Set of Mechanical Seals	1	1	2	2	3	3	3	
4	Set of Impeller Wear Rings	1	1	2	3	3	3	3	
5	Set of Casing Wear Rings	1	1	2	3	3	3	3	
6	Set of Gaskets, O - rings	2	4	6	8	8	8	8	
7	Complete set of coupling of each type	1	1	1	1	1	1	1	
	Seal Oil Pump Rotary (Main + Stand By)								
1	Complete bare pump assembly including driver couplings & base plate	1	1	1	1	1	2	2	

ITEM	SPARE PART	Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)							
		1	2	3	4	5	6	7	Notes
2	Set of bearings	1	1	2	3	3	3	3	
3	Set of Mechanical Seals	1	1	2	2	3	3	3	
4	Set of Gaskets, O - rings	2	4	6	8	8	8	8	
5	Complete set of coupling of each type	1	1	1	1	1	1	1	
	Emergency Oil Centrifugal Pump								
1	Complete bare pump assembly excluding driver couplings & base plate	1	1	1	1	1	2	2	
2	Set of bearings	1	1	2	3	3	3	3	
3	Set of complete seal with sleeve and gland plate	1	1	2	2	3	3	3	
4	Set of Impeller Wear Rings	1	1	2	3	3	3	3	
5	Set of Casing Wear Rings	1	1	2	3	3	3	3	
6	Set of Gaskets, O - rings	2	4	6	8	8	8	8	
7	Complete set of coupling of each type	1	1	1	1	1	1	1	
	Steam Turbine for Lube Oil Pump								
1	Dynamically balanced turbine Rotor	1	1	1	1	1	2	2	
2	Set of Bearings	1	1	2	3	3	3	3	
3	Set of Complete Shaft Seal Assembly	1	1	2	3	3	3	3	
4	Set of Gaskets, O-rings	2	4	6	8	8	8	8	
	Steam Turbine for Seal Oil Pump								
1	Dynamically balanced turbine Rotor	1	1	1	1	1	2	2	
2	Set of Bearings	1	1	2	3	3	3	3	
3	Set of Complete Shaft Seal Assembly	1	1	2	3	3	3	3	
4	Set of Gaskets, O-rings	2	4	6	8	8	8	8	
	Lube Oil Filters								
1	Filter cartridge / element along with set of gaskets, O - rings	10	20	30	30	30	30	30	
	Seal Filters (Oil l Gas)								
1	Filter cartridge / element along with set of gaskets, O - rings for Oil Seals	10	20	30	30	30	30	30	
2	Set of Seal Gas filter elements for Gas Seals	5	10	15	15	15	15	15	
	Oil Coolers								
1	Set of Gaskets, O-Rings	2	4	6	8	8	8	8	
	Inter-stage Coolers								
1	Set of Gaskets, O-Rings	2	4	6	8	8	8	8	
	After Coolers								

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
1	Set of Gaskets, O-Rings	2	4	6	8	8	8	8	
	ELECTRICAL								
	LV motor (3Ph Motors >1 KW & <=160KW)								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block (including Terminal Box Cover with Screws)	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	
	HV (6.6/11KV) induction motors >160KW								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block / HT bushing	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	

B.14 Packaged, Integrally Geared, Centrifugal Compressor

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	MECHANICAL								1,2,3,4
	Compressor								
1	Set of High Speed rotors (including pinion, shaft & impeller)	1	1	1	1	1	1	1	
2	Set of Compressor Bearings	1	1	2	3	3	3	3	
3	Set of Shaft Seals & carbon rings	1	2	3	3	3	3	3	
4	Set of all Labyrinths	1	1	2	3	3	3	3	
5	Set of Gaskets, Packings, O - rings	2	4	6	8	8	8	8	
6	Set of air filters elements	2	4	6	8	8	8	8	
7	Damper/ IGV Seals 100% Actuator-1 set of each type	1	1	1	1	1	1	1	
8	Complete Set of Coupling of Each Type	1	1	1	1	1	1	1	
	Lube Oil Pump Centrifugal (Main + Stand By)								
1	Complete bare pump assembly excluding driver couplings & base plate	1	1	1	1	1	2	2	
2	Set of bearings	1	1	2	3	3	3	3	
3	Set of Mechanical Seals	1	1	2	2	3	3	3	
4	Set of Impeller Wear Rings	1	1	2	3	3	3	3	
5	Set of Casing Wear Rings	1	1	2	3	3	3	3	
6	Set of Gaskets, O - rings	2	4	6	8	8	8	8	
7	Complete set of coupling of each type	1	1	1	1	1	1	1	
	Lube Oil Pump Rotary (Main + Stand By)								
1	Complete bare pump assembly excluding driver couplings & base plate	1	1	1	1	1	2	2	
2	Set of bearings	1	1	2	3	3	3	3	
3	Set of Mechanical Seals	1	1	2	2	3	3	3	

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
4	Set of Gaskets, O - rings	2	4	6	8	8	8	8	
	Lube Oil Filters								
1	Filter cartridge / element along with set of gaskets, O - rings	10	20	30	30	30	30	30	
	Lube Oil Coolers								
1	Set of Gaskets, O-Rings	2	4	6	6	8	8	8	
	Interstage Coolers								
1	Set of Gaskets, O-Rings	2	4	6	6	8	8	8	
	After Coolers								
1	Set of Gaskets, O-Rings	2	4	6	6	8	8	8	
	ELECTRICAL								
	Motor (Compressor Driver)								
1	Set of Bearings (NDE + DE)	1	1	2	3	3	3	3	
2	Set of Cooling Fans	1	1	1	1	1	2	2	
	Seal Filters (Oil l Gas)								
1	Filter cartridge / element along with set of gaskets, O - rings for Oil Seals	10	20	30	30	30	30	30	
2	Set of Seal Gas filter elements for Gas Seals	5	10	15	15	15	15	15	
	Oil Coolers								
1	Set of Gaskets, O-Rings	2	4	6	8	8	8	8	
	Interstage Coolers								
1	Set of Gaskets, O-Rings	2	4	6	8	8	8	8	
	After Coolers								
1	Set of Gaskets, O-Rings	2	4	6	8	8	8	8	
	ELECTRICAL								5

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	LV motor (3Ph Motors >1 KW & <=160KW)								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block (including Terminal Box Cover with Screws)	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	
	HV (6.6/11KV) induction motors >160KW								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block / HT bushing	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	

B.15 Special Purpose Steam Turbine

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	MECHANICAL								1,2,3,4
	Steam turbine								
1	Complete turbine spare rotor dynamically balanced (with permanent storage container) plates, spacers, shrouding etc.	1	1	1	1	1	1	1	
2	Complete set of radial bearings	1	2	3	4	4	4	4	
3	Complete set of thrust bearings	1	2	3	4	4	4	4	
4	Thrust disc (if detachable type) with nut	1	1	1	1	1	1	1	
5	Complete set of Shaft Seals	1	2	3	3	3	3	3	
6	Set of all labyrinths seals for complete turbine including inter-stage	1	2	3	3	3	3	3	
7	Set of Caulking tools & tackles for labyrinth insertions	1	1	1	1	1	1	1	
8	Complete set of Gaskets, 'O' rings	2	4	6	8	8	8	8	
	Condensate Extraction Pump - Centrifugal (Main + Stand By)								
1.	Set of bearings	1	1	2	3	3	3	3	
2	Set of complete seal with sleeve and gland plate)	1	1	2	2	3	3	3	
3.	Set of Impeller Wear Rings	1	1	2	3	3	3	3	
4.	Set of Casing Wear Rings	1	1	2	3	3	3	3	
5.	Set of Gaskets, O – rings	2	4	6	8	8	8	8	
	Turbine Cooling Oil Pump - Centrifugal (Main + Stand By)								
1.	Set of bearings	1	1	2	3	3	3	3	
2.	Set of Mechanical Seals	1	1	2	2	3	3	3	
3.	Set of Impeller Wear Rings	1	1	2	3	3	3	3	
4.	Set of Casing Wear Rings	1	1	2	3	3	3	3	
5.	Set of Gaskets, O – rings	2	4	6	8	8	8	8	
	Turbine Cooling Oil Pump- Rotary								
1.	Complete bare pump assembly excluding driver couplings & base plate	1	1	1	1	1	2	2	
2.	Set of bearings	1	1	2	3	3	3	3	
3.	Set of Mechanical Seals	1	1	2	2	3	3	3	
4.	Set of Gaskets, O – rings	2	4	6	8	8	8	8	
	Steam Turbine for Condensate Extr. Pump								

1	Dynamically balanced turbine Rotor	1	1	1	1	1	2	2	
2	Set of Bearings	1	1	2	3	3	3	3	
3	Set of Complete Shaft Seal Assembly	1	2	3	4	4	4	4	
4	Set of all Gaskets, O-rings	2	4	6	8	8	8	8	
	Surface Condensor								
1.	Set of gaskets, O – rings	2	4	6	8	8	8	8	
	Gland Steam Condensor								
1.	Set of gaskets, O – rings	2	4	6	8	8	8	8	
	Inter-Condensor								
1	Set of Gaskets, 'O' Rings	2	4	6	8	8	8	8	
	After-Condensor								
1	Set of Gaskets, 'O' Rings	2	4	6	8	8	8	8	
	Lube Oil Pump, Pump- Driver, Filters & Coolers								
1	Combined with driven equipment (Mandatory Spare Parts as applicable for the driven equipment shall be followed).								
	ELECTRICAL								
	LV motor (3Ph Motors >1 KW & <=160KW)								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block (including Terminal Box Cover with Screws)	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	
	HV (6.6/11KV) induction motors >160KW								
1	Set of Bearings (NDE + DE)	1	1	2	2	3	3	3	
2	Terminal Block / HT bushing	1	1	1	1	2	2	2	
3	Set of Cooling Fans	1	1	1	1	1	2	2	

B.16 Gear Box (Special / General Purpose):

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	MECHANICAL								1,2,3
1	Set of High Speed/Low Speed Gears & Shaft Assembly	1	1	1	1	1	1	1	
2	Set of Radial Bearings	1	2	3	4	4	4	4	
3	Set of Thrust Bearings	1	2	3	4	4	4	4	
4	Set of Labyrints, Oil Seals	1	2	3	4	4	4	4	
5	Set of all Gaskets, O-rings	2	4	6	8	8	8	8	
6	Complete bare lube oil pump (if integral with gear box)	1	1	1	1	1	1	1	

B.17 General Purpose Steam Turbine

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	MECHANICAL								1,2,3
1	Dynamically balanced turbine Rotor	1	1	1	1	1	2	2	
2	Set of Bearings	1	1	2	3	3	3	3	
3	Set of Complete Shaft Seal Assembly	1	2	3	4	4	4	4	
4	Set of all Gaskets, O-rings	2	4	6	8	8	8	8	
	Lube Oil Pump, Pump-Driver, Filters & Coolers								
1	Combined with driven equipment (Mandatory Spare Parts as applicable for the driven equipment shall be followed).								

B.18 Submersible pumps:

ITEM	SPARE PART	Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	MECHANICAL								1,2,3
	Pump								
1	Set of Impellers	1	1	2	3	3	3	3	
2	Set of Mechanical Seal	1	1	2	3	3	3	3	
3	Set of bearings	1	1	2	3	3	3	3	
4	Set of O-rings and Gaskets	2	2	3	3	4	4	4	

B.19 Air operated double diaphragm pump

ITEM	***SPARE PART***	***Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)***							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**Notes**
	MECHANICAL								1,2,3
1	Set of Diaphragms	2	2	2	3	3	3	3	
2	Set of Valve ball	2	2	2	3	3	3	3	
3	Set of valve seat	2	2	2	3	3	3	3	
4	Set of Shaft seal	2	2	2	3	3	3	3	
5	Set of Pilot sleeve assembly	2	2	2	3	3	3	3	

General Notes: These notes shall be applicable for items as mentioned under Section B: Mechanical – Rotary.

1. The word ‘Set' means the quantity required for full replacement of that part in one machine.

2. The CONTRACTOR shall quote for all the mandatory spares as defined above & as applicable to the proposed design of the equipment. In case, any spare which is listed above but not applicable due to specific construction/design of the equipment, the same shall be highlighted as ‘Not Applicable' against that spare supported with proper technical explanation, during the bidding stage.

3. Spare parts shall be identical in all respects to the parts fitted on the main equipment, including dimensions, material of construction & heat treatment.

4. Mechanical seal spares shall consist of –
   - Mechanical seal faces i.e. Seal ring and mating ring (including bellows if applicable)
   - Shaft sleeve
   - O rings / Springs for seal
   - Throttle bush

Section C: Mechanical – Package

C.1 FILTER

Item No.	Spare Part	Quantity Required (Based on the number of installed items)	Notes
1	Gaskets and Nozzles	100%	1
2	Bolts and Nuts for nozzle	5% (min. 2 sets)	1
3	Bolts and nuts for internals	5% (min. 2 sets)	
4	Filter Elements of disposal type	100% min.	

Note:-

1. For manholes, blinded flanges and companion flanges, when supplied.

C.2 DEAERATOR

Item No.	Spare Part	Quantity Required (Based on the number of installed items)	Notes
1	Gaskets and Nozzles	100%	1
2	Bolts and Nuts for nozzle	5% (min. 2 sets)	1
3	Bolts and nuts for internals	5% (min. 2 sets)	

Note:-

1. For manholes, blinded flanges and companion flanges, when supplied.

C.3 EJECTOR / EDUCTOR

Item No.	Spare Part	Quantity Required (Based on the number of installed items)	Notes
1	Ejector Single Stage, Nozzle (or Nozzle Components)	1 (1 Set)	1,3
2	Ejector Multi Stage, Nozzle (or Nozzle Components)	1 Per Stage (1 Set)	1,3
3	Eductor Single Stage, Nozzle (or Nozzle Components)	1 (1 Set)	2,3

Note:-

1) Ejector, Venturi or Venturi liner, not usually spared unless advised by vendor as wear prevalent. Remedial rework of Venturi or Venturi liner is possible but preferably carried out by specialist contractor.

2) Eductors, Chamber, Venturi or Venturi liner, each application to be spared. High wear is usually combated by using wear resistant materials e.g for abrasive solids transfer, ceramic chamber liner are used.

3) Ejectors, Eductors, Where multiple common units, same size, duty etc are used remains with single spare for nozzle Venturi or Chamber unless vendor advises wear prevalent review wear advice.

C.4 CRANE / ELECTRIC HOIST

Item No.	Spare Part	Quantity Required (Based on the number of installed items)	Notes
1	Brake Liner (long travel, cross travel, Hoist)	200%	
2	Bearing	200%	
3	Oil seals and Gaskets	200%	

C.5 AGITATOR

Item No.	Spare Part	Quantity Required (Based on the number of installed items)	Notes
1	Complete set of bearings for each agitator/mixer	1	
2	Mechanical seal cartridge for each agitator/mixer	1	
3	Shaft Sleeve for each agitator/mixer	1	
4	Oil seal for each agitator/mixer agitator	1	
5	Mechanical seal components (seal rings O-rings, springs) for each agitator/mixer	1	

C.6 COOLING TOWER

Item No.	Spare Part	Quantity Required (Based on the number of installed items)	Notes
1	PVC Fills	10%	
2	Nozzles	1 set	

C.7 MARINE / TANKER LOADING ARM

Item No.	Spare Part	Quantity Required (Based on the number of installed items)	Notes
1	Swivel Joints seal	1 set for each	
2	Seal for ERC	1 set	
3	Proximity switch	1 set	
4	Solenoid valve	1 set	
5	Filter Cartridge for hydraulic pump	1	

C.8 BOG PACKAGE

Item No.	Spare Part	Quantity Required (Based on the number of installed items)	Notes
1	Refrigerant charging pump, if applicable	1	
2	Lubricant oil pump, if applicable	1	
3	Oil filters	1	
4	Gaskets and O ring for compressor	1 (1 set)	
5	Mechanical shaft seal for compressor	1 (1 set)	
6	Bearing for compressor	1 (1 set)	
7	Gasket for vessel	2 sets of gaskets for each type and size	
8	Fasteners for vessels	10%(minimum 2 in each size) of installed fasteners	
9	Gaskets for Heat Exchanger	400%	
10	Fasteners for Heat Exchanger	20% load cell (rounded upto next higher digit)	

C.9 AIR SEPARATION UNIT

Item No.	Spare Part	Quantity Required (Based on the number of installed items)	Notes
1	Dynamically balanced rotor assembly	1	
2	Bearings	1 no. for each type and size	
3	Seals	2 no. for each type and size	
4	Gaskets	4 no. for each type and size	
5	O-rings	4 no. for each type and size	
6	Retainer rings	1 no. for each type and size	
7	Oil filter elements	4 no. for each type and size	

C.10 For the following items, mandatory spares shall be supplied by the CONTRACTOR as per VENDOR's recommendation for a swift operation purpose.

a) Waste Heat Boiler
b) Electric Heater
c) Plate Heat Exchanger
d) Barreling facility and
e) Any other item which is Vendor specific.

General Note:

1) For sub item or part list ie for eg "Motor"– Contractor shall follow the philosophy as mentioned in other Section.

2) In addition to the above, CONTRACTOR is also required to refer the spare philosophy, as mentioned in other sections.

3) For inconsistencies if any, most stringent requirement shall prevail.

Section D: Mechanical – Piping

ITEM	***SPARE PART***	***QUANTITY REQUIRED (% of As Built)***
1	Pipe and Fittings(½" to 1-1/2")	2
2	Pipe and Fittings(2" to 8")	2
3	Pipe and Fittings(10" to 36")	2
4	Valve and Flanges(½" to 1-1/2")	2 (Note-5)
5	Valve and Flanges(2" to 8")	2
6	Valve and Flanges(10" to 36")	2
7	Gaskets (except compressed asbestos type)	100
8	Bolts & Nuts	30 (Note-5)
9	Steam Traps(ALL SIZES)	2(Note-5)
10	Expansion Bellows (ALL SIZES)	100%
11	Miscellaneous items. (Hose, Special Couplings)	2(Note-5)
Note	01 number of spare requirement for Pipe and Fittings or for valves above 36 inch.	

NOTES:

(1) % of quantity to be supplied is against each & every item / size / material consumed in As-Built including piping.

(2) As a minimum one or one pipe length (6 Mtr Min) per item spare shall be supplied.

(3) For rating 900# & above & sizes 8" above, minmum one valve surplus shall be supplied for applicable sizes & rating, material.

(4) No spare quantity is required for process traps & motor operated valves.

(5) As a minimum one piece or one set.

Section E: Electrical

SL. NO.	PART DESCRIPTION	Quantity Required (Based on the number of installed items)
A.	**Transformer(one set of spares for each rating)**	
1	Bushing of each type with metal parts	1 set
2	H V	1 set
3	M V	1 set
4	Neutral	1 set
5	Complete set of gasket	1 set
6	Magnetic oil level gauge with 1 set of contacts	3 nos.
7	Oil temp. indicator with A/T contacts	2 nos.
8	Double float buchholz relay	1 nos.
9	Pressure relieving device	1no.
10	Silica gel breather	1 nos.
11	Valves	One of each type
12	Winding temp. indicator with A/T contacts	1 nos.
13	Neutral CT	1 set
14	Set of fuses of each rating for MB cubicles.	10 nos. for each rating
B.	**H V Switchboard (One set of spares per switchboard)**	
1	Load managers	1 no. for each bus
2	Spring charging motor	3 nos.
3	Closing coils	3 nos
4	Shunt Trip coil	for each rating 2 nos.
5	Bus support insulator	5 nos.
6	Control fuses (all types & ratings)	10 nos. for each rating
7	Indicating lamps	10 no for each colour
8	Limit switch for spring charging motor	2 nos.
9	Breaker control switch	2 nos.
10	Auxiliary switch	2 nos.
11	Push buttons	2 nos.
12	Anti-pumping contactor	2 nos.
13	Meter of each type	1 No of each type.
14	Vacuum bottles	5 nos. each rating

SL. NO.	PART DESCRIPTION	Quantity Required (Based on the number of installed items)
15	CT and PT (all types and ratings)	One no for each type & rating
16	All protection relays	One relay of each type
17	Panel limit switches and interlocking switches	5 nos.
18	Panel operating switches (all types)	One no of each type
19	Male pads	One set each for all types
20	Breaker limit switches & interlocking switches	5 nos.
21	Panel gasket	1 set
22	Panel door keys	2 nos
23	Control terminal blocks	15 nos. of each type
C	**M V Switchgear & MCC (One set of spares per switchboard)**	
1	ACB: (following spares shall be provided)	
2	Arcing contact (fixed)	1set of each rating
3	Arcing contact (moving)	1set of each rating
4	Spring charging motor	1set of each rating
5	6 way SIC block	1set of each rating
6	Panel limit switches & interlocking switches	5 of each type
7	Power / control fuse links (all types & ratings)	10 of each type & rating
8	Contactors	1 no for each rating
9	Overload relays	1 no for each rating
10	Push button unit	5 nos
11	Indicating lamps	10 nos. of each colour
12	Control terminal block	10 of each type
13	Meter of each type	1 No. of each type
14	Bus insulators	10 nos.
15	CT/ PT/Control transformer (all types and ratings)	3 nos.
16	Closing coils	2 nos.
17	Shunt trip coil	2 nos.
18	All protection -relays	One no. of each type
19	Male pads	One set each for all types
20	Breaker limit switches & interlocking switches	2 nos.

SL. NO.	PART DESCRIPTION	Quantity Required (Based on the number of installed items)
21	Gasket	One set
22	Door keys	2 nos.
23	Control connectors	2 set each for all types
24	Load managers	One for each bus
D.	**Lighting Fixtures (One set of spares for each type of lighting fixture)**	
1	Spare glass assembly for well glass fixtures	5 nos.
2	Neoprene gasket	5 nos.
3	Control gear components like capacitor, fuses etc.	1 set of each rating
4	Bulbs	10 of each rating
E.	**UPS (One set of spares for each rating of UPS)**	
1	SCR of all types & ratings	2 set of each rating
2	IGBT of all types & ratings	2 set of each rating
3	PCBs of all types	1 set
4	Control fuses	10 for each rating
5	Power fuses	10 for each rating
6	Switching module with static switch	1 set
7	Power supply card	1 set
8	Operating console card /control logic card	1 set
9	Cooling fans	1 set
F.	**Variable Frequency Drives (One set of spares for each rating of VFD)**	
1	Rectifier Control Module	1 set
2	Thyristor module	1 set
3	Inverter control unit	1 set
4	IGBT driver card	1 set
5	IGBT	2 nos. for each rating
6	PCB of each type	1 no.
7	Control fuse	10 nos. for each rating
8	Power fuse	10 nos.for each rating
9	Control cards	One set
10	Power supply card	One set

SL. NO.	PART DESCRIPTION	Quantity Required (Based on the number of installed items)
11	Relays	1 no. for each type of rating
12	Contactors	5 nos. for each type of rating
13	CT/PT	One no. for each type & rating
14	Cooling fans	One set of each type
15	Limiters & protection devices	One set
G.	**BATTERY CHARGER (One set of spares for each type & rating of Battery CHARGER)**	
1	Bridge diode for float charger	1 no. for each rating of SCR
2	Bridge diode for float cum boost charger	1 no. for each rating of SCR
3	Potentiometer	2 nos
4	Filter condenser	2 nos
5	Semiconductor HRC fuses for float charger	10 nos. for each rating of fuses
6	Semiconductor HRC fuses for float cum boost charger	10 nos. for each rating of fuses
7	Battery tap circuit blocking diode	2 nos.
8	Load break contactor	1 no.
9	Contactor coil	1 no.
H.	**BATTERY (One set of spares for each type & rating of Battery)**	
1	Cell container with cover per set	2 nos.
2	Vent plugs per set	2 nos
3	Acid level indicator(float) per set	2 nos
4	Inter cell connector per set	2 nos
5	Nut, bolts and washer per set	2 nos
I.	**SYNCHRONOUS GENERATOR**	
1	Rotating diodes	06 Nos for each generator
2	Power Supply unit for exciter	01 No for each generator
3	Power Supply unit for AVR	01 No for each generator
4	Transducer of each type	01 No for each generator
5	Metering and protection CTs	01 No of each type for each

SL. NO.	PART DESCRIPTION	Quantity Required (Based on the number of installed items)
		generator
6	LA for generator	01 No for each generator
7	Space heater	01 No for each generator
8	Bearing set with labyrinth rings	01 No for each generator
J.	**66kV GIS**	
1	Closing Coils	3 nos
2	Shunt Trip Coil	for each rating 2 nos.
3	Indicating lamps	10 no for each colour
4	CT	One no. for each type & rating
5	PT	One no. for each type & rating
6	All protection relays	One relay of each type

Section F:– Instrumentation

ITEM	*SPARE PART*	**Quantity Required (Based on the number of installed items)**
1	FIELD INSTRUMENTS	
1.1	Pressure Gauges/Differential pressure Gauge (Normal/Diaphragm type/Capillary type/Seal type)	20% (Subject to min. of 2) of each type, range, material of construction and rating.
1.2	Draft Gauges	10% (subject to minimum of 2) of each range of Draft gauge.
1.3	Temperature Gauges (Normal as well as Capillary)	20% (Subject to min. of 2) of each type / range of Temperature Gauges.
1.4	Level Gauges	A) For transparent gauges, 20% of illuminators with holder and reflector and 50% of bulbs
		B) In case of magnetic type level gauges, 10% of bi-color rollers for each gauge to be provided in addition to above.
		C) 30% subject to minimum one number of glass of each type, size along with pair of Gaskets (Cushion & Wet Gaskets),
		D) 20% subject to minimum one number of Gaskets for Gauge Cocks/Valves Bonnet / Packing
1.5	SMART Transmitters for Pressure, Flow (DP), Level(DP) and Temperature	10% (subject to minimum of 1) of each type, range, make and material of construction.
1.6	3 Way manifold	10% (subject to minimum of 1) of each size and material of construction.
1.7	Displacer Type Level Transmitters	a) Displacer and chain for each transmitters-10% (subject to minimum of 1)
		b) 10% (subject to minimum of 1) of Head assembly for each type, range and make ,
		c) 20% (subject to minimum of 1) of torque tube, pull rod, gasket spindle for mechanical zero adjustment.
1.8	Servo Type Level Gauges	A) 10% or minimum one number of each type of electronic card.
		B) 10% or minimum one number of each type of displacer.

		C) 10% or minimum one number of local indicators.
1.9	Ultrasonic / Radar instrument	A) 10% or minimum one number for each type of flow instrument & 10% or minimum one number probe / antenna for each level instrument
		B) 10 % or minimum one number of electronic card of each type
		C) 2 number fuses against each instrument

1.10	Capacitance/RF type instruments	10% or minimum one number of each type of electronic cards of each type.
1.11	Temperature Element: Thermocouple	20% (subject to minimum of 1) of each type, range, length of Thermocouple with Thermo well.
1.12	RTDs	20% (subject to minimum of 1) of each length of RTD with Thermowell.
1.13	Skin & flexible and multipoint bed thermocouples	10% (subject to minimum of 1) of each length & type of skin Thermocouple.
1.14	Ceramic Thermowell	1 no. for each type range, length.
1.15	Temperature transmitters	10% (subject to minimum of 1) of each type, range, make.
1.16	Variable Area Flowmeter (Rotameters)	20% or minimum one no. (Complete set) for each type, size, rating and material.
1.17	Averaging Pitot Tube	Set of Gasket, O-ring, Packing for Retract Mechanism and one no. Needle Valve with each Pitot Tube.
1.18	Mass Flow Meter (Coriolis and Thermal)	A) Power fuses 6 no. per set B) 10 % of Sensor Assembly or minimum 1 one.
		C) 10% or minimum one number of each type of electronic cards D) 10 % or minimum 1 No. of Display module.
1.19	Vortex Flow Meter	A)One set of gasket and Packing for each type and Size
		B) 10% or minimum one number of each type of electronic cards

1.20	Magnetic Flow Meter	a) One set of Gasket and Packing for each type and size. b) 10% or minimum one number of each type of electronic cards. Any additional items as mandatory spares if recommended by manufacturer.
1.21	PH & conductivity Transmitter	A) 10% or minimum two number of each type of sensor
		B) 10% or minimum one number of full set of transmitter
1.22	Nucleonic Level Instrument	10% or minimum one number of each type cards for detector and converter.
1.23	Neutron Back scattering Level Instrument	10% or minimum one number of each type cards for sensor and converter.
1.24	Copper tube	10% of installed quantity of each size.
1.25	SS tube	10% of installed quantity of each size
1.26	Pre insulated Tube	10% of installed quantity of each size
1.27	Instrument Signal Cable (single pair & multi pair)	10% of installed quantity of each size
1.28	Thermocouple extension cable	10% of installed quantity of each size
1.29	Control cable	10% of installed quantity of each size
1.30	Fibre Optic cable	10% of installed quantity of each size
1.31	Terminal blocks	10% of installed quantity of each size
1.32	Steam Trap	10% of installed quantity of each size
1.33	I/P Converter	10% of installed quantity
1.34	Fittings/Lugs/Nipples/ Sockets/blinds/Unions	10% of installed quantity of each type and size
1.35	½" Valves (Ball, Needle, Gate, Globe)	10% of installed quantity of each type
2	IN-LINE INSTRUMENTS	
2.1	Control Valves	A) 10 % or minimum 1 nos of each type, size, rating of Trim set consisting of seat, seat ring / seal ring, plug with stem, cage (wherever applicable), packing material.
		B) 20% (subject to minimum of 2) of Positioners (Smart) with links of each type and make.
		C) 20% (subject to minimum of 2) of Diaphragm.
		D) 20% (subject to minimum of 2) of Air filter regulators of each make.

		E) 10% (subject to minimum of 1) of Bonnet Gaskets/gland packings, piston O-rings, bearing & liner(of butter-fly valves) (for all types of valves)
		F) 10% (subject to minimum of 1) of any special accessories provided along with the control valve like boosters, position transmitters (wherever applicable), proximity switches, I/P converters etc.
2.2	Shutdown Valves	A) 20% (subject to minimum of 1) of Air filter regulators.
		b) Seal kit consisting of O rings, gaskets, etc. against each valve.
		c) 20% (subject to minimum of 1) of proximity switches.
		d) 20% (subject to minimum of 2) of solenoid valves of each type.
		e) 10% (subject to minimum of 1) of accessories like quick exhaust valves and temperature dependent fuses.
2.3	Pressure Relief Valves	a) Disc – 10% of total quantity of identical pressure relief valves or minimum one number whichever is higher.
		b) Expansion Bellows - 20% of total quantity of identical pressure relief valves or minimum one number whichever is higher
		c) Spring - 10% of total quantity of identical pressure relief valves or minimum one number whichever is higher
		d) Gasket Set - 20% of total quantity of identical pressure relief valves or minimum one number whichever is higher
2.4	Pressure control Valve (Self actuating valve)	A) Repair kit consisting of (orifice, plug, spring, gasket, diaphragm, O-ring for each valve.
		B) 20 % or minimum 2 nos of each type/size,rating etc. of Trim set consisting of seat, seat ring / seal ring, plug with stem, cage (wherever applicable), packing material for each valve to be provided as spare.

2.5.1	PRDS & De-superheater unit – (Manufactured in India)	A) 20% (subject to minimum of 2) of Positioners (Smart) with links of each type and make.
		B) 20% (subject to minimum of 2) of Diaphragm,
		C) 20% (subject to minimum of 2) of Air filter regulators of each make,
		D) 10% (subject to minimum of 1) of Bonnet Gaskets/gland packings, piston O-rings.
		E) 10% (subject to minimum of 1) of any special accessories provided along with the control valve like boosters, position transmitters (wherever applicable), proximity switches, I/P converters etc.
2.5.2	PRDS & De-superheater unit – (Manufactured outside India)	Same as those of manufactured in India and additionally: A) Spray Nozzle: 1 no of each size, type & rating
2.6	Butterfly valves	a)20% or minimum two set of actuator seal kits
		b) Same as that of Control valve
2.7	Saunder's Valves	20% or minimum two numbers of bottom body diaphragm
		20% (subject to minimum of 2) of Actuator diaphragms of Saunder valve of each type/size used.
2.8	Rupture Disc	As per manufacture's Recommendation
3	ANALYSER	(Spares for each analyser)
3.1	O2 Analyser	a)Furnace Assembly – 1
		b) Thermocouple –1
		c) Sensor board –1
		d) Each type/rating of Fuses – 1 set
		e) Thermistor assembly –1
		f) Ceramic filter –1
		g)Processor card – 1
		h)Power Supply card-1
		i)Display Board-1
		j)Cell Replacement kit-1
		k) Any Other card – 1

		L) Calibration kit (Regulator for Calibration gas cylinder and calibration rotameter) - One set for each of the units where O2 Analyser is installed.
3.2	Methane Analyser (Infrared)	a)Rings and gasket – 2 set of each type
		b) Sensor and Detector – 1 no.
		c) Rotameter – 1no.
		D) Pressure Control Valve – 1no.
		e)Fuses – 5 sets
		f) Electronic card – 1no.
		g)Other Aux. Cards – 1 each
		h) Processor card and other electronic – 1no. Each
3.3	Moisture Analyser	a) Sensor and detector –1 no.
		b) Electronic card – 1 no. each type
		c) Fuses – 5 sets
		d) Flow indicator & switch – 1 no.
3.4	Hydrogen Analyser	a)Rings and gasket – 2 set of each type
		b)Sensor and Detector – 1 no. with each
		c) Rotameter – 1no.
		d) Pressure Control Valve – 1no.
		e)Each type/rating of Fuses – 5 sets
		f) Electronic card – 1no. with each analyser
		g)Other Aux. Cards – 1 each
3.5	H2S Analyser	a) Cadmium lamp – 2 nos.
		b)Cell window – 2 nos.
		c)Window gasket – 10 nos.
		d) Air regulator – 1no.
		e)Seal kit fro ball valve – 4nos.
		f)Kit-o-ring – 2 sets
		g) Seal kit, Air act. – 2no.
		h)Fuse – 5 Set each
		I)Relay – 1 set each
		j)Display Board – 1no.with each analyser
		K) Processor Board –2no.
		l) Other Cards – 2each
3.6	Sulphur Analyser	a) Pyrolyzer Assembly- 1no.
		b) Thermocouple- 1 no.
		c) Heater Element- 1 no.
		d) Filter- 1 no.
		e) Valve(for sample)- no.
		f) Rotor
		g) O Rings- 100%
		h) UV lamps- 2 nos.

		i) Power supply module- 1 no.
3.7	CO + CO2 Analyser	a) Fuses – 1 set of each type
		b) Ring – 2 sets of each type
		c) Mother board –2 nos.
		d) Filter assembly band pass with guard – 2
		e) Filter – 1 set of each type
		f) IR/UV Lamp – 2 nos.
		Other cards – 1 each
3.8	Sox/Nox Analyser	a) Filter of each type – 2 sets
		b) Critical orifice – 1 no.
		c) Vacuum pump / Ejector – 1 no.
		d) Ring set – 1 no. each
		e) Processor / Mother board – 2 nos.
		f) Display board – 2 nos.
		g) Optical bench – 1 no.
		h) Filter detector – 2 nos.
		i) Fuses – 1 set of each type
		j) Other cards –1 each
		k) Ozonaotor –1
		l) Cuvette –2 nos
		m) chopper motor – 1 no.
		n) Dessican & flow sensor – 1 no. each
3.9	UV Analyzer	10% subject to minimum 1 No. of each type of modules.
	Mass Spectrometer	10% subject to minimum 1 No. of each type of modules.
3.10	Ethylene Analyzer	a)Rings and gasket – 2 set of each type
		b) Sensor and Detector – 1 no.
		c) Rotameter – 1no.
		D) Pressure Control Valve – 1no.
		e)Fuses – 5 sets
		f) Electronic card – 1no.
		g)Other Aux. Cards – 1 each
		h) Processor card and other electronic – 1no. Each
3.11	Chromatograph	a) Filter. – 2 sets
		b) Display Board –1 no.
		c) Processor Board –1 no.
		d) Chromatograph Column
		e) Solenoid valve (10% subject to minimum of 1 of each type
		f) Vaporization system if required, which includes vaporizer, thermostat, electrical tracing cable and heater etc. – 1 set

		g) Cooling system if required, which includes one cooler, flow conditioning system etc., - 1 set.
		h) Probe Filter – 2 Nos.
		i)Probe Tube – 2 Nos.
		J) Probe heater & Temp. sensors – 2 nos.
3.12	Sampling systems	Manufacturer's / Vendor's recommendation need to be followed.
3.13	SILICA ANALYSER	Control module- one
		Display board-10% or minimum one
		Stream selector-10% or minimum one
		Mixing chamber-10% or minimum one
		Bulb-10% or minimum one
		Pressure transmitter-10% or minimum one
		SOVs-10% or minimum 1
		Mirror-10% or minimum one
		Stirrer motor-10% or minimum one
		Cell kit and maintenance kit etc.10% or minimum two sets.
3.14	Dissolved Oxygen Analyser	a)Maintenance Kit-20% or minimum of two sets
4	DCS , PLC and SCADA	These spares are in addition to 20% installed spares
	A) Modules	10% or minimum two module of each type which are being used in DCS,PLCSER,DBM,DON,UHN and other system oriented items like I/O cards, all type of processor cards, all type of power supply cards, controlle cards, communication cards, interface cards etc. as required for the system including in consoles ,i.e. Operator /Engineering/PLC. This shall include spare modules/complete item for personal computer, PLC, Power supplies, LCD monitors/CRT, Disc Drives etc...
		10 % for items like hardwired instruments, barriers, signal converters, Trip amplifiers, auxiliary modules, relays, converters, annunciator cards, fuses, fused terminals, lamps/LEDs, Push buttons, selector switches.
		Prefabricated cables 10 Set with connecting plugs for each type of peripherals

		10 % or minimum 2 nos. of IS mV/I Converters, IS RTD/I Converters, Alarm cards (receiver switches)
		10% or minimum 2 nos of RS 232/485
		10% or minimum 2 nos of Keyboard / mouse / Track Balls
		System Cabinet Air Filter 100%
		10 % or minimum 2 nos. of Cabinet Cooling Fans of each type.
		20% or minimum one no. of each type of Fuses, MCBs, Auxiliary relays.
		10 % or minimum 2 Nos. of bulk power supply unit of each rating
		10 % or minimum 2 nos. of communication system components such as hubs, switches, routers, etc.
	B)Operator/Engineering Consoles PC/Subsystem PC	20% or minimum 2 numbers of latest version to be supplied as spare for each control room
	C) Consumables	Consumable spares including Laser printer cartridges, Blank CD (RW), Printer head Inking arrangement for hard copy unit Magnetic cartridge tape Floppy disc Printer Ribbon Paper or any other consumable items supplied along with the system required for a minimum of 6 months duration after system acceptance.
	D) Others	a) 10% or minimum two nos. of printers of each type
5	OTHER ITEMS	
5.1	Snubber, Syphon, Gauge Saver, etc.	10% (subject to minimum of 1) of each item used,
5.2	Loop powered indicators	20% (subject to minimum of 2) of Loop powered indicators used,
5.3	Solenoid Valves	20% or minimum 2 nos. of solenoid valves of each type used.

5.4	MCT Block	10% of each type of MCT Block used.
5.5	Panel mounted instruments	10% or minimum one number of each instrument
5.6.1	Flame Detector	10% subject to minimum one no. for each type
5.6.2	Beacon, Hooter	10% subject to minimum one no.
5.7	Annunciators	Power Supply Module 5% subject to minimum 1 No.of each type
		Tone Generator Card 5% subject minimum 1 No.of each type
		Flasher Module 5% subject to minimum 1 No.of each type
		I/O Module 5% subject to minimum 1 No.of each type - Lamps 10% - Fuses 1 dozen per Annunciator
		Hooter –5% or minimum one
5.8	CCTV	Camera - 20% (subject to minimum of 2) of each type
		Lens with zoom - 20% (subject to minimum of 2) of each type
		Receiver and drive - 20% (subject to minimum of 2)of each type
		Keypad including joystick - 20% (subject to minimum of 2) of each type
		All type of cards/modules - 20% (subject to minimum of 2) of each type
6	Machine Monitoring System/Condition monitoring system	A) 10 % or minimum 2 nos of each type of probes (vibration, speed, displacement etc.) and transducers/proximate/extension cables/connector protective sleeves etc.
		B) Electronic modules / cards : 10% or minimum 1 no. of each type,
		C) portable, offline measurement kits : 1 set of each type used
		D) monitor units & LCD display unit for local control panel : 10% or minimum 1 no. of each type,
		E) TDxNet cards : 1 set of each type used

		F) One Calibration kit for vibration probes for each unit wherever vibration probes are installed.
7	Gas detection system	Sensors : 20% or minimum 2 sets of each type,
		monitor units : 10% or minimum 1 no. of each type,
		Electronic cards / modules : 10% or minimum of 1 set of each type used,
8	Fire Detection and Alarm system	Sensors : 20% or minimum 2 sets of each type,
		monitor units : 10% or minimum 1 no. of each type,
		Electronic cards / modules: 10% or minimum of 1 set of each type used,
9	Custody Transfer metering	As per vendor's recommendation
10	Weigh Bridges/In motion weigh bridges	As per vendor's recommendation for two years of Maintenance.
11	LPG bottling plant	a)LPG Filling gonge:-10% or minimum two number b) Other field /control room instruments as per the spare philosophy listed above. c) Any special instrument as per vendor's recommendation.
12	Package equipment-Gas Turbine	a)Control room spares in line with that of DCS/PLC b)Field Instruments (Transmitters / Switches, vibration probes, SOVs, Control valves, analysers, thermocouples etc . in line with field instrument philosophy c)special instruments, As Per Vendor's Recommendation
13	Package equipment-Compressors (Hydrogen/nitrogen/PSA/ Air/other)	a)Control room spares in line with that of DCS/PLC b)Field Instruments (Transmitters / Switches, condition monitoring, SOVs, Control valves ,analyzers , thermocouples etc . in line with field instrument philosophy c)special instruments ,As Per Vendor's Recommendation

14	All other package equipments-Air driers or others	a)Control room spares in line with that of DCS/PLC b)Field Instruments (Transmitters /Switches,vibration probes,SOVs,Control valves,analysers,thermocouples etc . in line with field instrument philosophy c)special instruments ,As Per Vendor's Recommendation

Section G: Mechanical – Fire-Fighting

Sr. No.	Items	Quantity Required (BASED ON THE NUMBER OF INSTALLED ITEMS)
1.	Landing Valve / Hydrant Valve (Single Headed)	5% of Total Qty.
2.	Lug for Hydrant Valve	10% of Total Qty.
3.	MS Hand wheel for Hydrant Valve	10% of Total Qty.
4.	Spindle for Hydrant Valve as per SS 6603	10% of Total Qty.
5.	Seating for Hydrant Valve as per SS 304	10% of Total Qty.
6.	Rubber Seat Ring for Hydrant Valve	10% of Total Qty.
7.	Coupling Washer for Hydrant Valve	10% of Total Qty.
8.	Fire Hose Pipe with M/F Coupling	10% of Total Qty.
9.	Universal Triple Purpose Nozzle	10% of Total Qty.
10.	Air Release Valve	10% of Total Qty.
11.	Standard Spare Parts for Deluge Valve as per Deluge Valve manufacturer	1 set for each size of Deluge Valve.
12.	MVW Water Spray Nozzle	10% of Nozzle of each K-Factor
13.	HVW Water Spray Nozzle	10% of Nozzle of each K-Factor
14.	Quartzoid Bulb Detector	10% of Total Qty.
15.	Pressure Gauge	1 No. of each Type
16.	Solenoid Valve	1 No. of each Type
17.	Pressure Transmitter	1 No. of each Type
18.	6 Kg DCP Fire Extinguisher	10% of Total Qty.
19.	9 Kg DCP Fire Extinguisher	10% of Total Qty.
20.	4.5 Kg CO_2 Fire Extinguisher	10% of Total Qty.
21.	75 Kg DCP Mobile Trolley Mounted (Wheeled Type) Fire Extinguisher	10% of Total Qty.
22.	Clean agent cylinder	Min. one or 5% of total cylinder, whichever is more, shall be supplied with filled in clean agent gas. These are in addition to 100% standby filled up cylinder.

23.	Automatic and Manual release system	One of each type/size used in system supplied.
24.	Cylinder valve with safety pressure Relief device	One of each type/size used in system Supplied.
25.	Clean Agent Nozzles	10% of total nozzles used in each size/type with minimum drill hole
26.	Clean agent release push buttons	10% of total used of each type etc.

Note: % of quantity to be supplied is against each & every item / size / material consumed in As-Built.

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